



Original research

Changes in subjective mental and physical fatigue during netball games in elite development athletes

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ABSTRACT

Objectives: To assess the magnitude of changes in, and relationships between, physical and mental fatigue pre-to-post match in elite development netballers.**Design:** Observational.**Methods:** Twelve female netballers (21.3 ± 2.9 years) competing in the Australian Netball League reported perceptual measures of mental and physical fatigue pre- and post-match on 12 separate competition occasions. Minutes played, to allow for calculation of weighted changes (ratings proportional to playing time), positional groups and performance analysis variables were also assessed.**Results:** Post-match ratings were higher ($p < 0.01$) than pre-match for both mental (pre: 31.02 ± 19.28 ; post: 44.73 ± 24.47) and physical fatigue (pre: 28.86 ± 15.37 ; post: 47.21 ± 24.67). The correlation coefficient between change in mental and physical fatigue ($r = 0.37$, $p < 0.01$) revealed a shared variance of 13.9%. Acute changes in mental fatigue were related to minutes played ($r = 0.32$, $p < 0.01$) as was change in physical fatigue ($r = 0.59$, $p < 0.01$). No differences in change in mental fatigue were found between positional groups ($p = 0.07$) though change in physical fatigue was higher for shooters compared to defenders ($p < 0.05$). Performance variables revealed no relationships with mental or physical fatigue, with the exception of turnover number with both post-match weighted physical ($r = -0.23$, $p < 0.01$) and weighted-change in physical fatigue ($r = -0.16$, $p < 0.05$).**Conclusions:** Mental and physical fatigue were found to increase across netball matches in elite development athletes. Mental fatigue emerged as a largely separate construct to physical fatigue.

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Practical implications

- A 60 min netball match significantly increased subjective mental and physical fatigue in elite development athletes.
- Whilst competitive athletes may display an increased resistance to MF, they are not able to avoid MF.
- Mental and physical fatigue likely differ in origin and should both be evaluated and managed separately by practitioners.

Abbreviations: ANL, Australian Netball League; KPI, key performance indicator; MF, mental fatigue; PF, physical fatigue; RPE, rating of perceived exertion; SD, standard deviation; SSG, small sided games; SSN, Suncorp Super Netball; VAS, visual analogue scale.

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1. Introduction

Fatigue is a complex, multifaceted phenomenon¹ defined as a reduced capacity for maximal performance² and an inability to complete a task that was once achievable within a recent time frame.¹ Numerous studies have examined athlete fatigue and its relationship with training load,¹ performance,³ and the optimisation of recovery strategies⁴; fatigue remains a closely monitored and carefully considered variable with respect to athlete performance, training periodisation and recovery.⁴

A component of fatigue that has attracted attention in recent years is mental, or cognitive fatigue.^{5,6} Mental fatigue (MF) is defined as a psychobiological state caused by prolonged periods of demanding cognitive activity^{7,8} and is determined by subjective, behavioural and/or physiological changes.⁵ While MF has been shown to impair endurance cycling, running and swimming

tasks,^{5,6} intermittent running performance,⁹ and also physical performance during small sided soccer games,¹⁰ the influence of MF on repeated bouts of high intensity efforts over extended periods in a natural competitive sport setting is yet to be examined.

Much of the research that has investigated the relationship between MF and athlete performance has found MF to negatively impact physical,^{9,11,12} technical^{12,13} and tactical skills.^{10,14} However, it is yet to be shown that MF is present in athletes during competition. Furthermore, few studies have examined more than one outcome with elite athletes in settings that closely resemble the demands of competition.⁶ Further, while some studies have attempted to replicate match demands through simulations or small sided games,^{13–15} there is a need to examine MF in sporting competition to inform better monitoring and the development of strategies that may manage its potential effect on athlete performance.⁶

It is known that whilst subjective, behavioural and/or physiological changes coincide with MF, changes in all three of these areas do not have to be necessarily identified for mental fatigue to be present.⁵ Previous research has revealed sensitive and robust objective and physiological indicators of MF. The use of these assessment methods, such as electroencephalography (EEG),¹⁶ functional magnetic resonance imaging (fMRI)¹⁷ and cerebral blood flow (CBF)¹⁸, are unfortunately limited in the applied sporting environment due to practical constraints.^{6,16} Behavioural markers, including impairment in reaction time and decision-making accuracy have been found to occur with MF in response to traditional mentally fatiguing tasks such as the Stroop, Flanker and Psychomotor Vigilance Task (PVT).^{18,19} Whether the specificity of such established behavioural indicators remains sensitive beyond such tasks and can be translated for use with common tasks undertaken in the applied elite sporting environment remains contentious and an area for future research.⁶ While identifying a practical, non-invasive means to objectively assess MF is of great importance,⁶ using subjective measures to assess MF can nonetheless advance understanding of the field.

Netball is a high-intensity, intermittent team sport, played on a rectangular 30.5 × 15.25 m court.²⁰ The sport involves seven players on court and an additional 3–5 reserve players competing over four 15-min quarters. There are restrictions on where players can move on the court; positions are generally clustered into three sub-groups; goalers (goal shooter and goal attack), centre-court (wing attack, centre and wing defence) and defenders (goal defence and goal keeper). Netball is experiencing rapid advancement in professionalism, and common with other team sports, physical fitness, technical skill execution and application of tactical approaches significantly influence success in the sport.²⁰ Given that MF has been shown to impair physical,^{9,11,12} technical^{12,13} and tactical^{10,14} aspects in other team sport activities, it is reasonable to assume that MF influences the same components in netball. Positional differences in typical activities and associated physical workload have been demonstrated in elite netball; aiding the implementation of position-specific training.²¹ Theoretically, positional differences in mental workload may also exist; variation in technical demands between positional groups have been reported²² and, due to differences in the main performance aim of each positional group, e.g. attack, transition or defence, the application of tactics, which demands cognitive capacity,^{16,23} may also vary. However, whether playing time and the differing positional involvements lead to variation in, and differences between, physical and mental fatigue during competition is yet to be explored. Performance analysis is commonplace in netball to evaluate tactical and technical performance components including notational analysis to identify critical events within competition.²² Despite this,

published research informing relationships between such performance indicators and both physical and mental fatigue is lacking.

The primary aim of this study was to examine potential changes in MF and physical fatigue (PF) pre-to-post-match with elite development netballers. The purpose was to advance understanding of MF in netball to inform improved player monitoring and the development of strategies that can manage the effect of MF on athlete performance. It was hypothesised that both subjective MF and PF would be increased post-match, comparative to pre. In addition, it was hypothesised that MF and PF would emerge as separate perceptual constructs.

2. Methods

Twelve elite development female netballers (21.3 ± 2.9 y, 180.3 ± 5.5 cm, 76.16 ± 7.37 kg, 1016.47 ± 193.42 m Yo-Yo IRT1)²⁴ competing in the Australian Netball League (ANL) participated in the present study as determined by convenience sampling and ANL squad size regulations. The ANL is the feeder competition to the Suncorp Super Netball (SSN), the premier professional netball league in Australia. Fourteen matches, each lasting 60 min, were played over seven competition rounds; two matches were played per round on consecutive days, separated by a minimum of 20 h. Round 1 of competition was used for familiarisation and data were collected on each competition day; thus 12 matches were included in analysis. The study was approved by an ethics committee of The University of Queensland.

Thirty minutes prior to and within thirty minutes following each match, players completed MF and PF visual analogue scales (VAS); these have been previously validated,²⁵ and used¹² in the MF and physical performance literature. The pen and paper VAS scales required participants to rate their levels, by drawing a single vertical line to reflect 'mental fatigue' and 'physical fatigue' on separate 100 mm horizontal lines, both anchored from 'none at all' (0) to 'maximal' (100). To improve metacognition,¹⁶ definitions of mental and physical fatigue were provided to participants and examples of 'none at all' and 'maximal'. Mental fatigue was explained as "a psychobiological state caused by prolonged periods of demanding cognitive activity"^{7,8} and examples of traditional and sport-specific mentally fatiguing tasks, and descriptors as previously reported by athletes²⁶ were provided. Physical fatigue was explained as "a reduced capacity for maximal performance",² or an "inability to complete a task that was recently achievable, solely due to physical tiredness" and examples of physical fatigue experienced during familiar conditioning and resistance training activities were provided. To encourage truthfulness and reduce the potential for social-influence, response bias¹⁶ VAS scales were completed in a quiet section of the team locker room with each individual providing subjective responses out of sight of other athletes, coaching or support staff. Participants were aware of the scales being used to monitor fatigue but were not specifically informed that MF was the primary focus of the research — the aim being to further reduce the potential for acquiescence bias.¹⁶ Key performance indicator (KPI) data during matches were recorded using SportsCode Elite (Hudl, 2019). For each match this included; score differential, centre pass conversion (%), turnover conversion (%), absolute turnover number and shooting accuracy (%).²² Minutes on court (playing time) and position were recorded. Positional groups were defined by standard categorisation for playing position; shooters (goal shooter and goal attack), centre-court (wing attack, centre and wing defence) and defenders (goal defence and goal keeper).

Following normality testing, Wilcoxon-signed rank tests were used to assess pre- post- changes in MF and PF, change in MF comparative to PF, and in-round, between-match differences (e.g.

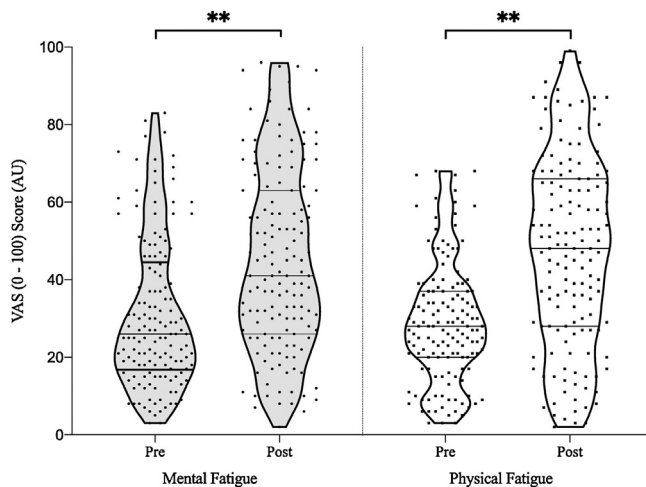


Fig. 1. Violin plot including individual data points for pre- and post-match VAS (AU) ratings of MF and PF: minimum, Q1, median, Q4 and maximum scores. ** $p < 0.01$ = statistically significant increase comparative to pre-ratings. Abbreviations: VAS, visual analogue scale; MF, mental fatigue; PF, physical fatigue.

round 4: day 1 vs day 2). Due to violation of normality assumptions, Kruskal Wallis (non-parametric ANOVA) tests were used to examine differences between positional groups for weighted-change in MF and PF. Weighted scores of MF and PF were calculated to represent responses proportional to playing time i.e. minutes of involvement in each match, using the formula; weighted subjective MF or PF = ((minutes played $\times$ subjective change)/total minutes of court time available). Spearman's rank correlational test determined the shared variance between acute changes in MF and PF and were used to establish the relationship between court time (i.e. number of minutes played) and ratings of MF and PF. Pearson's correlations were used to investigate relationships between weighted-post match and weighted-change in MF and PF subjective ratings and team performance analysis KPI variables. A subset of weighted-post match and weighted-change in MF and PF inclusive of 'shooters' ratings only were used to assess relationships with shooting accuracy. Pearson's correlation coefficients were also used to investigate the relationships between score differential (how much a match was won or lost by) and pre-, post- and change in ratings of MF and PF. Statistical analyses were performed using R Studio software (R Foundation, V.1.1.423), Excel (Microsoft, 2016) and visualised using Prism (GraphPad, V.8.10). Alpha was set at 0.05.

3. Results

As shown in Fig. 1, both MF and PF significantly changed between pre- and post-match ($p < 0.01$). Differences between pre- and post-match ratings of MF and PF, acute changes in MF comparative to PF and in-round, between-match differences for change in, pre-ratings of MF and PF are shown in Table 1.

There was a significant difference in shared variance between change in MF and PF ($r = 0.37$, $p < 0.01$); the proportion of shared variance between change in MF and change in PF as two ranked variables was thus 13.9%. Acute changes in MF were significantly related to minutes played (time on court) ($r = 0.32$, $p < 0.01$); acute change in PF were also significantly related to minutes played ($r = 0.59$, $p < 0.01$).

There were no significant differences found for change in MF between positional groups ($p = 0.07$) and no differences identified for PF between shooters and centre-court players ($p = 0.43$), or between centre-court and defensive players ($p = 0.57$). A significantly higher change in PF was found for shooters compared to

Table 1

Wilcoxon Signed rank statistics for MF and PF comparisons.

VAS comparison	Mean $\pm$ SD	p-Value
Pre-match MF v	31.02 $\pm$ 19.28	
Post-match MF	44.73 $\pm$ 24.47	$<0.01^{**}$
Pre-match PF v	28.86 $\pm$ 15.37	
Post-match PF	47.21 $\pm$ 24.67	$<0.01^{**}$
Acute change in MF v	13.71 $\pm$ 22.15	
Acute change in PF	18.23 $\pm$ 23.83	0.112
Day 1 pre-match MF v	29.26 $\pm$ 17.60	
Day 2 pre-match MF	32.83 $\pm$ 20.84	0.412
Day 1 pre-match PF v	26.37 $\pm$ 14.82	
Day 2 pre-match PF	31.39 $\pm$ 15.60	$<0.05^{*}$

Abbreviations: VAS, visual analogue scale; MF, mental fatigue; PF, physical fatigue.

* $p < 0.05$.

** $p < 0.01$.

defenders ($p < 0.05$). During the season, the average match score differential was -12.3 goals; with 3 wins and 11 losses reported. No significant relationships were found between score differential and; pre-match MF ($r = 0.07$, $p = 0.41$), post-match MF ($r = 0.03$, $p = 0.72$), change in MF ($r = -0.03$, $p = 0.74$), pre-match PF ($r = -0.05$, $p = 0.57$), post-match PF ($r = 0.09$, $p = 0.25$), or change in PF ($r = 0.13$, $p = 0.13$). Table 2 shows the relationships between performance analysis (KPI) variables and ratings of fatigue.

4. Discussion

The aim of the study was to assess changes in, and the relationship between, mental and physical fatigue during competition with elite development netball players. Subjective measures of both MF and PF significantly increased during a 60 min netball match, however changes in each were not strongly related. The mean pre-post-changes and post-match ratings (44.7 ± 24.5) of MF found in the present study are similar to those reported by Veness et al.¹¹ in elite male cricketers following 35 min of exposure to the incongruent Stroop Test (46.6 ± 5.9 AU). The wider standard deviations found in the present study, compared to Veness et al., are explainable by the likelihood that in the competition setting, participants were exposed to broader levels of cognitive demands over the duration of the match. This is plausibly due to differences in playing and substitution time, or varied levels of cognitive challenge due to the quality of their direct positional opponent. Conversely, in the cross-over design study,¹¹ participants followed the same protocol thus narrower deviations from the mean are seen. Similarly, the post-match ratings in the present study liken to those found from research assessing MF and technical performance in well-trained soccer players following the Stroop task, 58 ± 22 AU.¹² The present post-match ratings were however lower than those found with the same well-trained soccer players when they undertook a modified (increased difficulty) 30 min version of the incongruent Stroop Test (79.5 ± 13.3 AU).²⁷ Conversely, in research with 20 Australian National Premier League soccer players where Badin et al. assessed mental and physical fatigue responses with small sided games (SSG)¹⁵; the ratings of MF in this study (63 ± 20 AU) were consistent with those found following a 30 min computerised Stroop task and a 15 min SSG (65 ± 11 AU), undertaken following the Stroop.¹⁵ The exact physiological mechanisms accompanying MF remain unclear, and technology is not yet able to directly assess changes in cerebral adenosine concentrations,²⁸ or phosphocreatine and glucose availability.¹⁹ Further research should however aim to assess subjective, behavioural and physiological measures to compare MF experienced during competition to MF resulting from mentally fatiguing tasks such as the Stroop or AX-CPT in the same group of athletes. The present data show that subjective VAS ratings of MF following a netball match are comparable to data derived from other studies following mentally fatiguing tasks and

Table 2

Pearson's correlations showing relationship of MF and PF with both team and shooter KPI metrics.

Fatigue	KPI	p-Value	Pearson's r
Team metric			
Post-match weighted MF	Centre pass conversion (%)	0.47	0.06 moderate
	Turnover conversion (%)	0.43	0.07
	Absolute turnover #	0.09	–0.14
Post-match weighted PF	Centre pass conversion (%)	0.07	0.15
	Turnover conversion (%)	0.25	0.09
	Absolute turnover #	<0.01**	–0.23
Weighted change MF	Centre pass conversion (%)	0.75	0.03
	Turnover conversion (%)	0.06	–0.05
	Absolute turnover #	0.96	0.00
Weighted change PF	Centre pass conversion (%)	0.30	0.09
	Turnover conversion (%)	0.20	0.11
	Absolute turnover #	<0.05*	–0.16
Shooter (GS, GA) metric			
Post-match weighted shooter MF		0.76	0.05
Post-match weighted shooter PF	Shooting	0.36	–0.14
Weighted-change in shooter MF	accuracy (%)	0.93	0.01
Weighted-change in shooter PF		0.36	–0.14

Abbreviations: MF, mental fatigue; PF, physical fatigue.

* p < 0.05.

** p < 0.01.

that the contributors to MF during competitive netball outweigh the potentially contradictive benefits of exercise and arousal, also experienced during a match.²⁹ Accordingly, the findings align with previous claims that whilst competitive athletes may display an increased resistance to MF,³⁰ they are not able to avoid MF.³¹

The present data show that while there is some overlap, MF and PF are largely different constructs. Whilst statistically significant, analysis found a low proportion of variance in change in MF to be explained by variance in PF, with 86.1% remaining unexplained. Previous research has shown MF to negatively influence physical performance and this has been primarily attributed to an increased perception of effort as a result of adenosine accumulation and decreased dopamine concentrations in the brain.²⁸ Accordingly, the performance decrements are attributed to an origin different to skeletal muscle,²⁸ or that typically are considered as physical fatigue. This concept is broadly supported by differential ratings of perceived exertion (RPE) scores of breathlessness, leg muscle exertion and technical/cognitive exertion RPE scores; with higher technical/cognitive exertion displayed by athletes when playing higher ranked opposition in soccer.³² The relationships between MF and PF to minutes played (time on court) in the present study also support this concept to warrant future exploration and are of practical interest. Despite both demonstrating significance, MF demonstrated a weaker relationship (explained variance) with minutes played compared to PF and minutes played. There are a number of potential explanations for these findings. Firstly, on-court players experience cognitive demands of response-inhibition, information recall, sustained vigilance and concentration, information switching and distraction to anticipation capacity.⁶ Though the physical demands of a substitute players are clearly lower, whilst on the interchange bench, such cognitive demands may still be experienced given that players remain visually and audibly engaged. Other additional mental demands may be present, such as analysis provided to team-mates, lack of cohesion from coaching staff and management of the disappointment of limited court time, requiring emotional regulation. These cognitive demands, experienced for a sustained duration,³¹ i.e. 60 min of a competitive netball match, may result in inducement of MF in substitute players and provide one explanation that the relationship between MF and playing time is weaker than that of PF. Accordingly, the current within-competition findings inform coaches, support staff and athletes that the resulting MF of being a substitute player, with limited court time is different to physical fatigue.

Another plausible explanation for the weaker relationship between time on-court and MF, is variation in the individual susceptibility to MF (as shown in Fig. 1).³³ Whilst the physical demands of sporting competition are reasonably well understood and these inform appropriate physical preparation undertaken by all athletes, mental training, with the aim to better equip players for the cognitive demands of competition is rarely implemented. Accordingly, with the appropriate physical preparation in each positional demand, players may experience similar physical fatigue responses to competition. A greater variability in the capacity to cope with cognitive demands of the competitive sporting environment may however exist. This concept of increased cognitive heterogeneity is further supported by the greater spread of pre-ratings of MF compared to PF (Fig. 1). As such, future research should continue to explore MF and PF responses both acutely across competition, and longitudinally to inform response norms and variability within athletic populations in the elite sporting environment. Furthermore, teams should consider the inclusion of specific mental training protocols to appropriately prepare players for the MF experienced during competition. 'Brain endurance training', such as undertaking moderate intensity activity e.g. at 65% maximal oxygen consumption, whilst also performing a cognitively demanding task, e.g. the AX-CPT,³⁴ may provide an avenue in which such adaptations can be achieved.

The present study revealed no differences in acute change in MF between positional groups and the only positional difference in PF was found between shooters and defenders. These findings are similar to soccer, where no positional differences in RPE-technical, RPE-breathlessness and RPE-leg muscle exertion score have been reported, with the exception of full-backs.³² Though differences in technical involvements between positions in netball, e.g. passes, number of intercepts and goal attempts, have been found,²² the present findings suggest that cognitive demands may not differ between positions in netball in elite development players.

Mental and physical fatigue have both previously been shown to impact multiple aspects of team sporting-performance when assessed in controlled environments; including execution of technical^{12,13} and tactical^{10,14} and physical^{9,11,12} variables. In soccer, subjective ratings of MF have been shown to negatively impact offensive and defensive technical variables during SSGs.¹⁵ Evidence also supports team dynamics, such as dyads synchronisation to be decreased with MF.¹⁴ Furthermore, MF negatively influenced open-skill visuomotor performance of badminton players.³¹ However, no

significant relationships were found between mental and physical fatigue and performance analysis variables in the present study, with the exception of turnover number with weighted-change in PF and post-match weighted PF. Despite a duration of 60 min, netball requires interspersed high-intensity efforts involving repetition of power-based movements of sprinting and jumping.²¹ As such, the inconclusive evidence demonstrating a potentially insignificant impact of MF on high-intensity or maximal tasks of short duration⁶ combined with the contextual sport specific demands of netball may provide logical explanation for the observed findings. It is recognised that potential limitations; reduced skill level of the elite development athletes,³⁵ subsequent likelihood of errors, and lack of statistical power, may have influenced the likelihood for detection of significant relationships. Further research should investigate MF, PF and performance variables in the competition environment; the inclusion of individual player involvements and performance ratings is recommended.

A strength and novel aspect of the present research is its applied nature; however, the limited opportunity to obtain mechanistic information to expand knowledge on the physiology behind MF is acknowledged. The collection of multiple subjective (VAS) scales, i.e. mental and physical fatigue, was undertaken with processes to improve metacognition and reduce the potential impact of acquiescence bias and demand characteristics.¹⁶ Despite the non-significant findings in the present study, the average negative score differential is acknowledged and a potential influence of match outcome on subjective ratings of mental fatigue remains an area for future research. Further, although findings may be generalised to team sports, consideration for the specificity to elite female netballers competing across 4 × 15 min quarters, should be taken with MF potentially sensitive to periods of rest and time-on-task.²²

In conclusion, the present study found subjective increases in MF and PF during competition with elite development athletes. In addition, MF was a largely separate construct to PF. Finally, no relationships were found between MF and performance measures during competition.

Authors' contributions

All authors were involved in writing, reviewing and editing the manuscript.

Ethical approval and consent to participate

The study was approved by the human ethics committee at The University of Queensland (approval number 2018000792). Informed consent was obtained from all individuals who participated in the study.

Consent for publication

On the consent form, it was explained to potential participants that we, the researchers, intended to submit the findings for publication. By signing, they agreed to this.

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